Seaborn

What is Seaborn?

Imagine you have a giant box of crayons, but every time you want to draw a graph, you have to measure every line with a ruler and mix the paints yourself. That's **Matplotlib**! Now imagine you get a **magic stencil paintbrush** that draws beautiful, colorful charts automatically in just 1 second. That magic paintbrush is **Seaborn**!

Why do python programmers love Seaborn?
- ■ ***Automatic Beauty:*** *Pretty default colors, clean grid lines, and crisp typography.*
- ■ ***One-Line Code:*** *High-level commands do complex statistical math for you.*
- ■ ***Data-Ready:*** *Plugs directly into Pandas DataFrames effortlessly.*

Getting Started: The Magic Setup

To start drawing with Seaborn, we load our tools in Python:

```
import seaborn as sns
import matplotlib.pyplot as plt

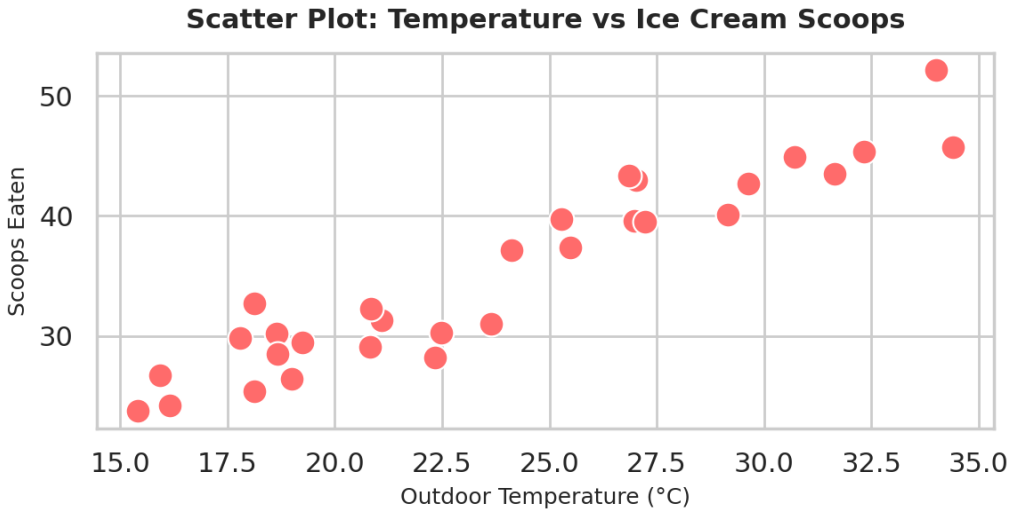
# Set a stylish default theme
sns.set_theme(style='whitegrid')
```

The 4 Main Plot Superpowers

1. Relational Plots: Connecting Dots

Scatter Plot (sns.scatterplot) shows if two things are related — like outdoor temperature and ice cream scoops sold!

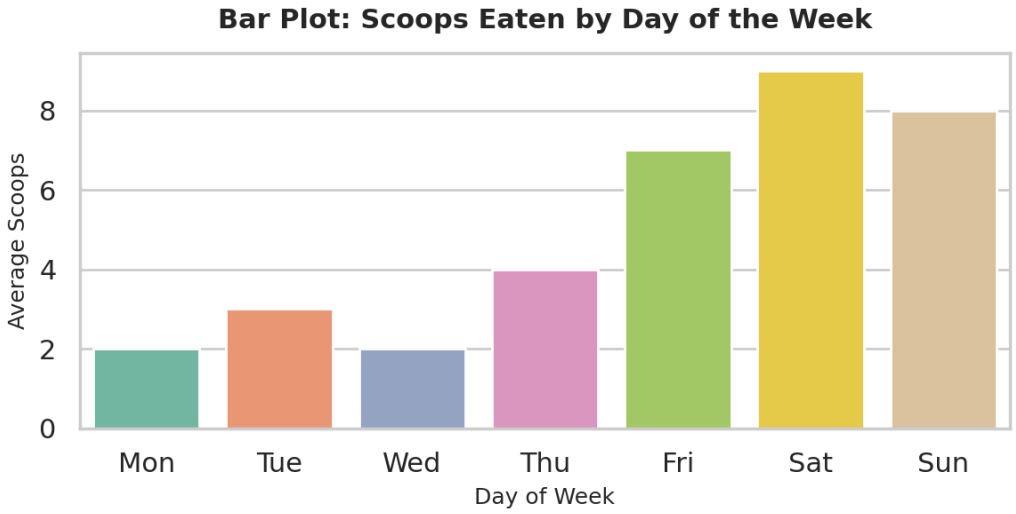
```
# Drawing a scatter plot
sns.scatterplot(x=temperature, y=ice_cream_scoops, color='#FF6B6B')
plt.title('Temperature vs Ice Cream Scoops')
plt.show()
```



2. Categorical Plots: Comparing Groups

Bar Plot (sns.barplot) compares quantities across different groups or days of the week.

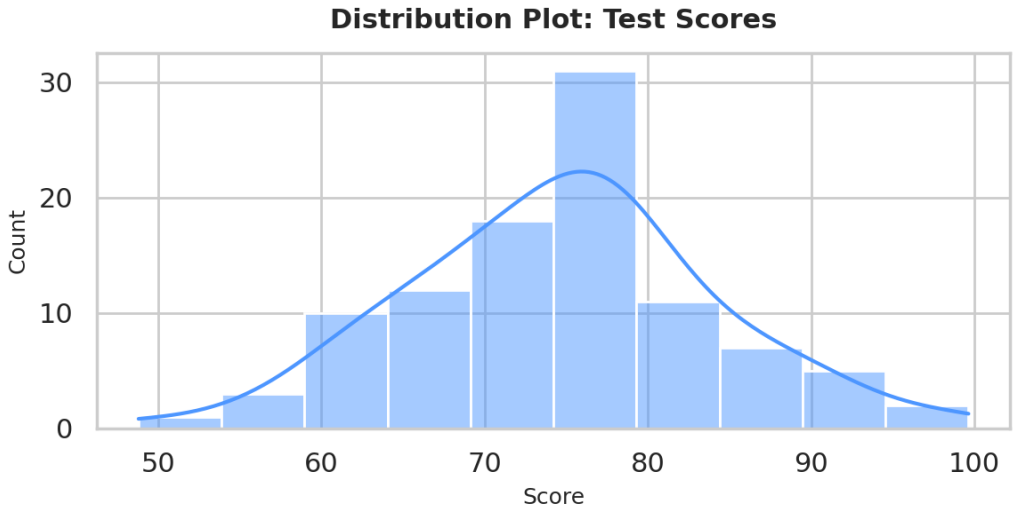
```
# Comparing scoops eaten across days of the week
sns.barplot(x=days, y=scoops, palette='Set2')
plt.title('Scoops Eaten by Day of Week')
plt.show()
```



3. Distribution Plots: Finding the Spread

Histogram & KDE (sns.histplot) shows how data values are grouped together and where most numbers fall.

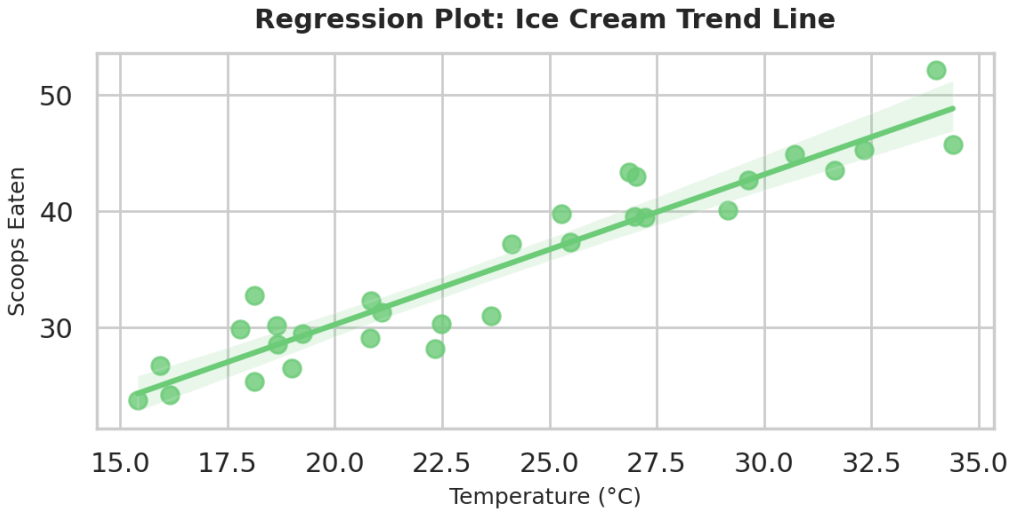
```
# Viewing test score distributions
sns.histplot(test_scores, kde=True, color='#4D96FF')
plt.title('Test Score Distribution')
plt.show()
```



4. Regression Plots: Spotting Trends

Regression Plot (sns.regplot) draws a smart trend line right through your data to make future predictions!

```
# Draw scatter points + a predictive trend line
sns.regplot(x=temperature, y=ice_cream_scoops, color='#6BCB77')
plt.title('Ice Cream Trend Line')
plt.show()
```



Quick Cheat Sheet

Plot Type	Function	What It Does	Kid Analogy
Scatter Plot	sns.scatterplot	Compares two continuous variables	Dot-to-dot puzzle
Bar Plot	sns.barplot	Compares amounts across groups	Stacking Lego towers
Histogram	sns.histplot	Shows spread & frequencies	Sorting marbles into buckets
Regression Plot	sns.regplot	Scatter plot with a trend line	Drawing a laser beam through dots